Award Criterion 1: Technical & Scientific Excellence

Specific Guidelines for the Technical Evaluator

Section 1: Scope of Technical Excellence

The Technical Evaluator assesses the scientific quality, novelty, methodological rigor, and technical feasibility of the proposed work.

The evaluation covers:

- **Clarity of Objectives**: Specific, measurable, achievable, relevant, and time-bound (SMART) objectives.
- **Soundness of Methodology**: Theoretical foundation, mathematical formalisms, machine learning model architectures, and validation benchmarks.
- **Novelty vs State of the Art**: Genuine advance beyond current frontiers rather than incremental packaging of existing libraries.
- **Consistency of Visual Artifacts vs Text**: High-level claims, abstracts, and executive summaries must strictly match accompanying figures, architectural schematics, and benchmark charts. Any discrepancy between text (e.g. "+40% efficiency boost") and actual graphical data (e.g. chart showing 12%) is considered a major methodological weakness.

Section 2: Technical Sub-Criteria

1. **Methodological Rigor**: Does the proposal explain training datasets, validation protocols, baseline comparisons, and ablation plans?
2. **Technical Feasibility**: Are computational resources, hardware access (GPUs, TPUs), and dataset licenses secured?
3. **Reproducibility & Open Science**: Code repository availability, adherence to FAIR data principles (Findable, Accessible, Interoperable, Reusable).